

Design Introduction

Study Guide

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1.5.1 JVM – Java Virtual Machine

- JVM (Java Virtual Machine) is an abstract machine.
- It is called a virtual machine because it doesn't physically exist.
- It is a specification that provides a runtime environment in which Java bytecode can be executed.
- It can also run those programs which are written in other languages and compiled to Java bytecode.
- JVM, JRE, and JDK are platform dependent because the configuration of each OS is different from each other. However, Java is platform independent.

✂ Responsibilities:

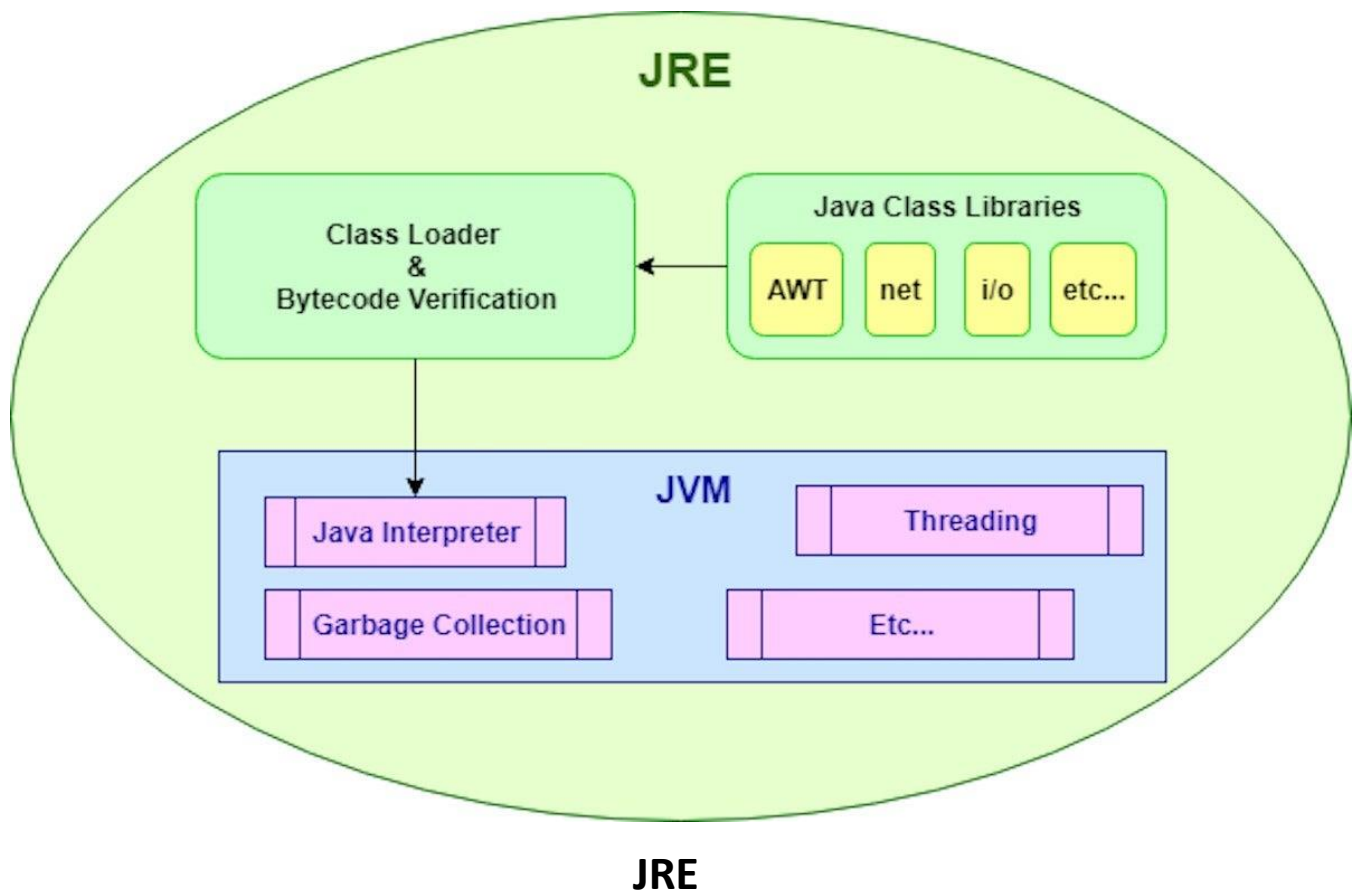
- Loads .class files.
- Verifies bytecode for security.
- Converts bytecode into machine code (via Interpreter or JIT Compiler).
- Manages memory (via Garbage Collection).
- Ensures exception handling and multithreading.

🔄 JVM Lifecycle:

1. Class Loader
2. Bytecode Verifier
3. Execution Engine
4. Runtime Memory Areas (Stack, Heap, Method Area)

1.5.2 Java Runtime Environment

- JRE is an acronym for Java Runtime Environment. It is also written as Java RTE. The Java Runtime Environment is a set of software tools which are used for developing Java applications. It is used to provide the runtime environment. It is the implementation of JVM. It physically exists. It contains a set of libraries + other files that JVM uses at runtime.
- The implementation of JVM is also actively released by other companies besides Sun Micro Systems.



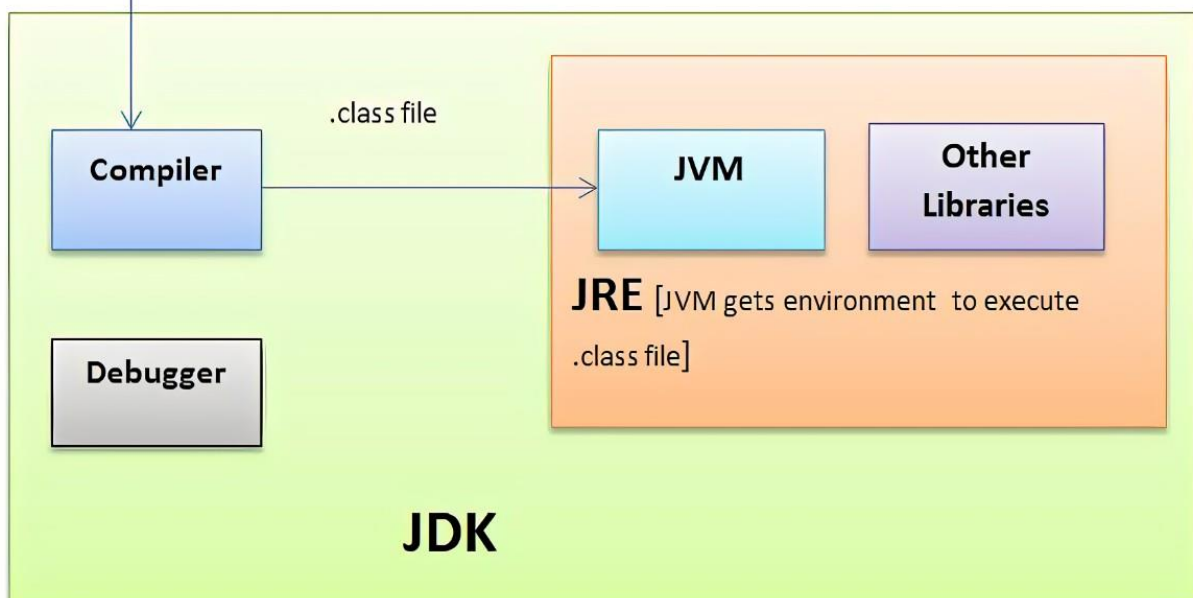
1.5.3. Java Development Kit

- JDK is an acronym for Java Development Kit. The Java Development Kit (JDK) is a software development environment which is used to develop Java applications and applets. It physically exists. It contains JRE + development tools.

Components:

- JRE (Java Runtime Environment)
- javac – Java compiler
- java – Launcher tool
- javadoc – Documentation tool
- jar – Archiving tool
- debuggers and other development tools

Source files (.java files)



➤ Summary of Key Concepts:

- **JVM:** Core engine that runs Java bytecode on any platform.
- **JRE:** Provides the runtime to **execute** Java programs.
- **JDK:** Full toolset to **develop and run** Java programs.

 **JDK = JRE + Development Tools**

 **JRE = JVM + Java Libraries**

➤ References:

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3. OpenJDK JVM Implementations

 <https://openjdk.org/>

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5. GeeksforGeeks – Java Virtual Machine (JVM) Architecture

 <https://www.geeksforgeeks.org/java-virtual-machine-jvm/>

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